

BRAINSTORM GROUP (BSG)

MAT111
(Pure Mathematics)
Solution to Quadratic Questions



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If the root of the equation $2x^2 + 5x + 3 = 0$ are α and β. Then obtain the equation whose root values are:

(a) α^2 and β^2

(b) $1/\alpha$ and $1/\beta$

(c) α^3 and β^3

(d) $1/\alpha^2$ and $1/\beta^2$

(e) $\alpha + 1/\beta$ and $\beta + 1/\alpha$

(f) β/α^2 and α/β^2

(g) $\alpha^2 + \beta$ and $\beta^2 + \alpha$

(h) $2\alpha + \beta$ and $2\beta + \alpha$

SOLUTION

From general equation:

$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$

Let $S_r \equiv$ sum of roots

$P_r \equiv$ product of roots

we have

$$x^2 - (S_r)x + (P_r) = 0$$

Since α and β are the roots!

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \quad -(*)$$

Comparing the equation above with equation: $2x^2 + 5x + 3 = 0$

we have

$$S_r: \alpha + \beta = -5/2$$

$$P_r: \alpha\beta = 3/2$$

(a) Given roots: α^2 and β^2.

$$S_r = \alpha^2 + \beta^2$$

$$(\alpha+\beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$$

$$(\alpha+\beta)^2 - 2\alpha\beta = \alpha^2 + \beta^2$$

$$S_r = (\alpha+\beta)^2 - 2\alpha\beta$$

$$= (-5/2)^2 - 2(3/2) = \frac{25}{4} - 3 = \frac{13}{4}$$

$$P_r = \alpha^2\beta^2 = (\alpha\beta)^2 = (3/2)^2 = \frac{9}{4}$$

The equation required is therefore

$$x^2 - (S_r)x + (P_r) = 0$$

$$x^2 - (13/4)x + (9/4) = 0$$

$$4x^2 - 13x + 9 = 0$$

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b.) Given roots: $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

$S_r = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-5/2}{3/2}$

$= -\frac{5}{3}$

$P_r = \frac{1}{\alpha} \times \frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{1}{3/2}$

$= \frac{2}{3}$

The required equation is:

$x^2 - (-\frac{5}{3})x + \frac{2}{3} = 0$

$3x^2 + 5x + 2 = 0$

c.) Given roots: $\alpha^3 + \beta^3$

$S_r = \alpha^3 + \beta^3$

$(\alpha+\beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$

$(\alpha+\beta)^3 = \alpha^3 + 3\alpha\beta(\alpha+\beta) + \beta^3$

$(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = \alpha^3 + \beta^3$

$S_r = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$

$= (-\frac{5}{2})^3 - 3(\frac{3}{2})(-\frac{5}{2})$

$= -\frac{125}{8} + \frac{45}{4} = -\frac{35}{8}$

$P_r = \alpha^3\beta^3 = (\alpha\beta)^3$

$= (3/2)^3 = \frac{27}{8}$

The required equation is:

$x^2 - (-\frac{35}{8})x + \frac{27}{8} = 0$

$8x^2 + 35x + 27 = 0$

d.) Given roots: $\frac{1}{\alpha^2}$ and $\frac{1}{\beta^2}$

$S_r = \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2+\beta^2}{\alpha^2\beta^2}$

from ⓐ: $\alpha^2 + \beta^2 = 13/4$

and $\alpha^2\beta^2 = 9/4$

$S_r = \frac{13/4}{9/4} = \frac{13}{9}$

$P_r = \frac{1}{\alpha^2} \times \frac{1}{\beta^2} = \frac{1}{\alpha^2\beta^2} = \frac{1}{9/4}$

$= \frac{4}{9}$

The required equation is:

$x^2 - (\frac{13}{9})x + \frac{4}{9} = 0$

$x^2 - 13x + 4 = 0$

e.) Given roots: $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$

$S_r = (\alpha + \frac{1}{\beta}) + (\beta + \frac{1}{\alpha})$

$= \frac{\alpha^2\beta + \alpha + \alpha\beta^2 + \beta}{\alpha\beta}$

$= \frac{\alpha\beta(\alpha+\beta) + (\alpha+\beta)}{\alpha\beta}$

$= \frac{(\alpha+\beta)(\alpha\beta+1)}{\alpha\beta} = \frac{(-\frac{5}{2})(\frac{3}{2}+1)}{\frac{3}{2}}$

$= \frac{(-5/2)(5/2)}{3/2} = \frac{-25/4}{3/2} = -\frac{25}{6}$

$P_r = (\alpha + \frac{1}{\beta})(\beta + \frac{1}{\alpha}) = \alpha\beta + 2 + \frac{1}{\alpha\beta}$

$= \frac{3}{2} + 2 + \frac{2}{3} = 25/6$

The required equation is:

$$x^2 - \left(-\frac{25}{6}\right)x + \frac{25}{6} = 0$$

$$\underline{6x^2 + 25x + 25 = 0}$$

f) Given roots: β/α^2 and α/β^2

$$S_r = \frac{\beta}{\alpha^2} + \frac{\alpha}{\beta^2} = \frac{\beta^3 + \alpha^3}{\alpha^2\beta^2}$$

from ⓒ: $\alpha^3 + \beta^3 = -\frac{35}{8}$

from ⓐ: $\alpha^2\beta^2 = \frac{9}{4}$

$$S_r = \frac{-35/8}{9/4} = -\frac{35}{18}$$

$$P_r = \frac{\beta}{\alpha^2} * \frac{\alpha}{\beta^2} = \frac{1}{\alpha\beta} = \frac{2}{3}$$

The required equation is:

$$x^2 - \left(-\frac{35}{18}\right)x + \frac{2}{3} = 0$$

$$\underline{18x^2 + 35x + 12 = 0}$$

g) Given roots: $\alpha^2+\beta$ and $\beta^2+\alpha$

$$S_r = (\alpha^2+\beta)+(\beta^2+\alpha)$$

$$= (\alpha^2+\beta^2)+(\alpha+\beta)$$

from ⓐ: $\alpha^2 + \beta^2 = \frac{13}{4}$

$$S_r = \frac{13}{4} + \left(-\frac{5}{2}\right) = \frac{3}{4}$$

$$P_r = (\alpha^2+\beta)(\beta^2+\alpha)$$

$$P_r = \alpha^2\beta^2 + \alpha^3 + \beta^3 + \alpha\beta$$

from ⓐ: $\alpha^2\beta^2 = \frac{9}{4}$

from ⓒ: $\alpha^3 + \beta^3 = -\frac{35}{8}$

$$P_r = \frac{9}{4} - \frac{35}{8} + \frac{3}{2} = -\frac{5}{8}$$

The required equation is:

$$x^2 - \left(\frac{3}{4}\right)x - \frac{5}{8} = 0$$

$$\underline{8x^2 - 6x - 5 = 0}$$

h) Given roots: $2\alpha+\beta$ and $2\beta+\alpha$

$$S_r = (2\alpha+\beta) + (2\beta+\alpha)$$

$$= 3\alpha + 3\beta = 3(\alpha+\beta)$$

$$= 3\left(\frac{-5}{2}\right) = -\frac{15}{2}$$

$$P_r = (2\alpha+\beta)(2\beta+\alpha)$$

$$= 4\alpha\beta + 2\alpha^2 + 2\beta^2 + \alpha\beta$$

$$= 2(\alpha^2+\beta^2) + 5\alpha\beta$$

from ⓐ: $\alpha^2 + \beta^2 = \frac{9}{4}$

$$P_r = 2\left(\frac{9}{4}\right) + 5\left(\frac{3}{2}\right) = \frac{9}{2} + \frac{15}{2}$$

$$= \frac{24}{2} = 12$$

The required equation is:

$$x^2 - \left(-\frac{15}{2}\right)x + \frac{24}{2} = 0$$

$$\underline{2x^2 + 15x + 24 = 0}$$

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Resolve the following in terms of S and P, where

$S = \alpha + \beta$ and $P = \alpha\beta$

(a) $\alpha^2 + \beta^2$ (b) $\alpha^3 + \beta^3$ (c) $\alpha^4 + \beta^4$ (d) $\alpha - \beta$

(e) $\alpha^2 - \beta^2$ (f) $\alpha^3 - \beta^3$ (g) $\alpha^3 - \beta^3$

SOLUTION

(a) $(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$

$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

$= S^2 - 2P$

(b) $(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$

$= \alpha^3 + 3\alpha\beta(\alpha+\beta) + \beta^3$

$\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$

$= S^3 - 3PS$

(c) $(\alpha+\beta)^4 = \alpha^4 + 4\alpha^3\beta + 6\alpha^2\beta^2 + 4\alpha\beta^3 + \beta^4$

$= \alpha^4 + 4\alpha\beta(\alpha^2+\beta^2) + 6(\alpha\beta)^2 + \beta^4$

$\alpha^4 + \beta^4 = (\alpha+\beta)^4 - 4\alpha\beta(\alpha^2+\beta^2) - 6(\alpha\beta)^2$

Recall from (a): $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

Hence

$\alpha^4 + \beta^4 = (\alpha+\beta)^4 - 4\alpha\beta[(\alpha+\beta)^2 - 2\alpha\beta] - 6(\alpha\beta)^2$

$= S^4 - 4P(S^2 - 2P) - 6P^2$

$= S^4 - 4PS^2 + 8P^2 - 6P^2$

$= S^4 - 4PS^2 + 2P^2$

(d) $(\alpha-\beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$

$= (\alpha+\beta)^2 - 2\alpha\beta - 2\alpha\beta$

$= (\alpha+\beta)^2 - 4\alpha\beta$

$= S^2 - 4P$

$\alpha - \beta = \sqrt{S^2 - 4P}$

(e) $\alpha^2 - \beta^2 = (\alpha+\beta)(\alpha-\beta)$

$= S\sqrt{S^2 - 4P}$

(f) $\alpha^3 - \beta^3 = (\alpha^2 + \alpha\beta + \beta^2)(\alpha-\beta)$

$= (\alpha^2 + \beta^2 + \alpha\beta)(\alpha-\beta)$

$= [(\alpha+\beta)^2 - 2\alpha\beta + \alpha\beta](\alpha-\beta)$

$= [(\alpha+\beta)^2 - \alpha\beta](\alpha-\beta)$

$= (S^2 - P)\sqrt{S^2 - 4P}$

(g) $\alpha^4 - \beta^4 = (\alpha^3 + \alpha^2\beta + \alpha\beta^2 + \beta^3)(\alpha-\beta)$

$= [(\alpha^3 + \beta^3) + \alpha\beta(\alpha+\beta)](\alpha-\beta)$

$= (S^3 - 3PS + PS)\sqrt{S^2 - 4P}$

$= (S^3 - 2PS)\sqrt{S^2 - 4P}$

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